Q1. JEE Main 2026 (21 January Shift 2)

If $\int_0^1 4\cot^{-1}\left(1-2x+4x^2\right)\mathrm{d}x = \mathrm{a}\tan^{-1}(2) - \mathrm{b}\log_e(5)$, where $\mathrm{a}, \mathrm{b} \in \mathbf{N}$, then $(2\mathrm{a}+\mathrm{b})$ is equal to ____.

Q2. JEE Main 2026 (23 January Shift 1)

The value of the integral $\int_{\frac{\pi}{24}}^{\frac{5\pi}{24}} \frac{\mathrm{d}x}{1+\sqrt[3]{\tan 2x}}$ is :

(1) $\frac{\pi}{3}$ (2) $\frac{\pi}{12}$ (3) $\frac{\pi}{18}$ (4) $\frac{\pi}{6}$

Q3. JEE Main 2026 (28 January Shift 2)

Let $[\cdot]$ denote the greatest integer function. Then $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{12(3+[x])}{3+[\sin x]+[\cos x]}\right)\mathrm{d}x$ is equal to :

(1) $13\pi+1$ (2) $12\pi+5$ (3) $11\pi+2$ (4) $15\pi+4$

Q4. JEE Main 2026 (24 January Shift 2)

If $f(x)$ satisfies the relation $f(x) = e^x + \int_0^1 (y+xe^x)f(y)dy$, then $e+f(0)$ is equal to ____.

Q5. JEE Main 2026 (22 January Shift 2)

Let $f(x) = [x]^2 - [x+3] - 3, x \in \mathbf{R}$, where [] is the greatest integer funtion. Then

(1) $\int_0^2 f(x)\mathrm{d}x = -6$

(2) $f(x) < 0$ only for $x \in [-1, 3)$

(3) $f(x) > 0$ only for $x \in [4, \infty)$

(4) $f(x) = 0$ for finitely many values of x

Q6. JEE Main 2026 (22 January Shift 2)

Let $[\cdot]$ be the greatest integer function. If $\alpha = \int_0^{64} \left(x^{1/3} - \left[x^{1/3}\right]\right)\mathrm{d}x$, then $\frac{1}{\pi}\int_0^{\alpha\pi} \left(\frac{\sin^2\theta}{\sin^6\theta+\cos^6\theta}\right)\mathrm{d}\theta$ is equal to ____ .

Q7. JEE Main 2026 (22 January Shift 1)

The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1}{[x]+4}\right)dx$, where $[\cdot]$ denotes the greatest integer function, is

(1) $\frac{1}{60}(21\pi-1)$ (2) $\frac{7}{60}(\pi-3)$

(3) $\frac{7}{60}(3\pi-1)$ (4) $\frac{1}{60}(\pi-7)$

Q8. JEE Main 2026 (23 January Shift 2)

The number of elements in the

set $\mathrm{S} = \left\{x : x \in [0, 100] \text{ and } \int_0^x t^2 \sin(x-t)\mathrm{d}t = x^2\right\}$ is ____.

Q9. JEE Main 2026 (21 January Shift 1)

The value of $\int_{-\pi/6}^{\pi/6}\left(\frac{\pi+4x^{11}}{1-\sin(|x|+\pi/6)}\right)dx$ is equal to:

(1) 8π (2) 6π (3) 2π (4) 4π

Q10. JEE Main 2026 (21 January Shift 1)

$6\int_0^{\pi}|(\sin 3x+\sin 2x+\sin x)|dx$ is equal to ____.

Q11. JEE Main 2026 (28 January Shift 1)

Let f be a polynomial function such that $f\left(x^2+1\right)=x^4+5x^2+2$, for all $x\in\mathbb{R}$. Then $\int_0^3 f(x)dx$ is equal to

(1) $\frac{5}{3}$ (2) $\frac{41}{3}$ (3) $\frac{27}{2}$ (4) $\frac{33}{2}$

Q12. JEE Main 2026 (28 January Shift 1)

If $\int\left(\frac{1-5\cos^2 x}{\sin^5 x\cos^2 x}\right)dx=f(x)+\mathrm{C}$, where C is the constant of integration, then $f\left(\frac{\pi}{6}\right)-f\left(\frac{\pi}{4}\right)$ is equal to

(1) $\frac{1}{\sqrt{3}}(26+\sqrt{3})$ (2) $\frac{1}{\sqrt{3}}(26-\sqrt{3})$

(3) $\frac{4}{\sqrt{3}}(8-\sqrt{6})$ (4) $\frac{2}{\sqrt{3}}(4+\sqrt{6})$

Q13. JEE Main 2026 (28 January Shift 1)

The value of $\sum_{r=1}^{20}\left(\left|\sqrt{\pi\left(\int_0^r x|\sin\pi x|dx\right)}\right|\right)$ is ____

Q14. JEE Main 2026 (24 January Shift 1)

Let a differentiable function f satisfy the equation $\int_0^{36} f\left(\frac{tx}{36}\right)dt=4\alpha f(x)$. If $y=f(x)$ is a standard parabola passing through the points $(2,1)$ and $(-4,\beta)$, then β^{α} is equal to ____.

ANSWERS AND SOLUTIONS

1. 9	**2.** (2)	**3.** (3)	**4.** 2	**5.** (2)	**6.** 36	**7.** (3)	**8.** 16
9. (4)	**10.** 17	**11.** (4)	**12.** (3)	**13.** 210	**14.** 64		

1. **(9)** Let $I = \int_0^1 \cot^{-1}(1-2x+4x^2)dx$

$I = \int_0^1 \left(\cot^{-1}(2x-1) - \cot^{-1}(2x)\right) dx$

Applying king

$I = \int_0^1 \left(-\cot^{-1}(2x-1) + \cot^{-1}(2x-2)\right) dx$

From (1) and (2)

$2I = \int_0^1 \left(\cot^{-1}(2x-2) - \cot^{-1}(2x)\right) dx$

$= \int_0^1 \cot^{-1}(2x-2)dx - \int_0^1 \cot^{-1}(2x)dx$

Applying King

$= \int_0^1 \cot^{-1}(-2x)dx - \int_0^1 \cot^{-1}(2x)dx$

$= \int_0^1 \left(\pi - \cot^{-1}(2x)\right) dx - \int_0^1 \cot^{-1}(2x)dx$

$= \int_0^1 \left(\pi - 2\cot^{-1}(2x)\right) dx$

$= \pi - 2\int_0^1 \left(\cot^{-1} 2x\right) \cdot 1dx$

By parts

$= \pi - 2\left[\left(x\cot^{-1} 2x\right)_0^1 + \int_0^1 \frac{2x}{1+4x^2}dx\right]$

Let $1 + 4x^2 = t$

$8xdx = dt$

$= \pi - 2\left[\cot^{-1} 2 + \frac{1}{4}\int_1^5 \frac{dt}{t}\right]$

$= \pi - 2\cot^{-1} 2 - \frac{1}{2}\ell n5$

$2I = 2\tan^{-1} 2 - \frac{1}{2}\ell n5$

$\Rightarrow 4I = 4\tan^{-1} 2 - \ell n5$

$\therefore 2a + b = 8 + 1 = 9$

2. **(2)** Let $I = \int_{\pi/24}^{5\pi/24} \frac{dx}{1+\sqrt[3]{\tan 2x}}$.

Apply King's rule ($x \to \pi/4 - x$):

$I = \int_{\pi/24}^{5\pi/24} \frac{dx}{1+\sqrt[3]{\cot 2x}} = \int_{\pi/24}^{5\pi/24} \frac{\sqrt[3]{\tan 2x}}{1+\sqrt[3]{\tan 2x}} dx.$

Adding: $2I = \int_{\pi/24}^{5\pi/24} dx = \frac{5\pi}{24} - \frac{\pi}{24} = \frac{\pi}{6}.$

$I = \frac{\pi}{12}.$

3. **(3)** $I = \int_{-\pi/2}^{\pi/2} \frac{12(3+[x])\,dx}{3+[\sin x]+[\cos x]}$

$I = \int_{-\pi/2}^{-1} \frac{12(1)\,dx}{2} + \int_{-1}^{0} \frac{12(2)\,dx}{2} + \int_{0}^{1} \frac{12(3)\,dx}{3} + \int_{1}^{\pi/2} \frac{12(4)\,dx}{3}$

$I = 6\left(\frac{\pi}{2}-1\right) + 12(0+1) + 12(1-0) + 16\left(\frac{\pi}{2}-1\right)$

$I = 3\pi - 6 + 12 + 12 + 8\pi - 16$

$I = 11\pi + 2$

4. **(2)** Given: $f(x) = e^x + \int_0^1 (y + xe^y) f(y)dy$

Let $A = \int_0^1 f(y)dy$ and $B = \int_0^1 ye^y f(y)dy$.

Then $f(x) = e^x + Ax + Be^x = (1+B)e^x + Ax$

Integrating both sides from 0 to 1:

$A = \int_0^1 [(1+B)e^y + Ay]dy = (1+B)(e-1) + \frac{A}{2}$

This gives $\frac{A}{2} = (1+B)(e-1)$, so $A = 2(1+B)(e-1)$

From $f(0) = 1 + B$ and solving the coupled equations for A and B:

$f(0) = 1 - e$

Therefore $e + f(0) = e + (1-e) = 2$

5. **(2)** For $x \in [n, n+1)$ where n is an integer, $[x] = n$ and $[x+3] = n+3$.

Thus $f(x) = n^2 - (n+3) - 3 = n^2 - n - 6 = (n-3)(n+2)$.

For $f(x) < 0$, we need $(n-3)(n+2) < 0$, which gives $-2 < n < 3$, i.e., $n \in \{-1, 0, 1, 2\}$.

This corresponds to $x \in [-1, 3)$.

We can verify:

For $n = -1$: $f = 1 - 2 - 3 = -4 < 0$

For $n = 0, 1$: $f = -6 < 0$

For $n = 2$: $f = 4 - 5 - 3 = -4 < 0$

For $n = 3$: $f = 9 - 6 - 3 = 0$

For $n \geq 4$, $(n-3)(n+2) > 0$.

Therefore $f(x) < 0$ only for $x \in [-1, 3)$.

6. **(36)** $\because \int_0^{64} x^{\frac{1}{3}}\,dx = \frac{3}{4}\left[x^{\frac{4}{3}}\right]_0^{64} = 192$ &

$\int_0^{64} [x^{1/3}]\,dx = \int_0^1 [x^{1/3}]\,dx + \int_1^8 [x^{1/3}]\,dx + \int_8^{27} [x^{1/3}]\,dx + \int_{27}^{64} [x^{1/3}]\,dx = 156$

So $\alpha = 192 - 156 = 36$

Now $E = \frac{1}{\pi} \int_0^{36\pi} \frac{\sin^2\theta}{\sin^6\theta + \cos^6\theta}\,d\theta$

$= \frac{36}{\pi}\int_0^{\pi}\frac{\sin^2\theta}{\sin^6\theta+\cos^6\theta}\,d\theta$

$\Rightarrow E = \frac{36\cdot 2}{\pi}\int_0^{\pi/2}\frac{\sin^2\theta}{\sin^6\theta+\cos^6\theta}\,d\theta$

Let $J = \int_0^{\pi/2}\frac{\sin^2\theta}{\sin^6\theta+\cos^6\theta}\,d\theta$ ....(1)

Applying King

$J = \int_0^{\pi/2}\frac{\cos^2\theta}{\sin^6\theta+\cos^6\theta}\,d\theta$ .....(2)

Now $2J = \int_0^{\pi/2}\frac{1}{\sin^6\theta+\cos^6\theta}\,d\theta$ (add (1) & (2))

$= \int_0^{\pi/2}\frac{\sec^6\theta}{\tan^6\theta+1}\,d\theta$

$= \int_0^{\infty}\frac{(1+\lambda^2)}{\lambda^4-\lambda^2+1}\,d\lambda$

$= \int_0^{\infty}\frac{1+\frac{1}{\lambda^2}}{\lambda^2-1+\frac{1}{\lambda^2}}\,d\lambda$

$= \pi$

$\Rightarrow J = \frac{\pi}{2}$

$\Rightarrow E = \frac{36\cdot 2}{\pi}\times J = 36$

7. **(3)** $I = \int_{-\pi/2}^{\pi/2}\frac{1}{[x]+4}\,dx$

$I = \int_{-\pi/2}^{-1}\frac{dx}{2} + \int_{-1}^{0}\frac{dx}{3} + \int_0^1\frac{dx}{4} + \int_1^{\pi/2}\frac{dx}{5}$

$= \frac{1}{2}\left(-1+\frac{\pi}{2}\right) + \frac{1}{3}(1) + \frac{1}{4}(1) + \frac{1}{5}\left(\frac{\pi}{2}-1\right)$

$= \frac{7\pi}{20} - \frac{7}{60} = \frac{7}{60}(3\pi-1)$

8. **(16)** Using $u = x - t$: $\int_0^x t^2\sin(x-t)\,dt = \int_0^x (x-u)^2\sin u\,du$.

Expanding and integrating term by term:

$= x^2(1-\cos x) - 2x(-x\cos x + \sin x) + (-x^2\cos x + 2x\sin x + 2\cos x - 2)$

$= x^2 + 2\cos x - 2$.

Setting equal to x^2: $2\cos x - 2 = 0 \Rightarrow \cos x = 1 \Rightarrow x = 2n\pi$.

In $[0, 100]$: $2n\pi \le 100 \Rightarrow n \le \frac{50}{\pi} \approx 15.9$.

So $n = 0, 1, 2, \ldots, 15$, giving 16 elements.

9. **(4)** Split: $\int_{-\pi/6}^{\pi/6}\frac{\pi}{1-\sin(|x|+\pi/6)}dx + \int_{-\pi/6}^{\pi/6}\frac{4x^{11}}{1-\sin(|x|+\pi/6)}dx$.

Second integral = 0 (odd function over symmetric interval).

First integral: $2\pi\int_0^{\pi/6}\frac{1}{1-\sin(x+\pi/6)}dx$.

Let $u = x + \pi/6$: $= 2\pi \int_{\pi/6}^{\pi/3} \frac{1}{1-\sin u} du$.

$\frac{1}{1-\sin u} = \sec^2 u + \sec u \tan u$.

$= 2\pi[\tan u + \sec u]_{\pi/6}^{\pi/3} = 2\pi[(\sqrt{3}+2) - \sqrt{3}] = 4\pi$.

10. (17) Using $\sin 3x + \sin x = 2\sin 2x \cos x$:

$\sin 3x + \sin 2x + \sin x = \sin 2x(2\cos x + 1) = 2\sin x \cos x(2\cos x + 1)$

Zeros in $[0, \pi]$: $x = 0, \pi/2, 2\pi/3, \pi$

Expression is positive on $(0, \pi/2)$, negative on $(\pi/2, 2\pi/3)$, positive on $(2\pi/3, \pi)$

Let $F(x) = -\frac{\cos 3x}{3} - \frac{\cos 2x}{2} - \cos x$

$F(0) = -\frac{11}{6}$, $F(\pi/2) = \frac{1}{2}$, $F(2\pi/3) = \frac{5}{12}$, $F(\pi) = \frac{5}{6}$

$\int_0^{\pi/2} = \frac{7}{3}$, $\left|\int_{\pi/2}^{2\pi/3}\right| = \frac{1}{12}$, $\int_{2\pi/3}^{\pi} = \frac{5}{12}$

Total $= \frac{7}{3} + \frac{1}{12} + \frac{5}{12} = \frac{17}{6}$

$6 \times \frac{17}{6} = 17$

11. (4) Let $u = x^2 + 1$, then $x^2 = u - 1$.

Substituting into the given condition:

$f(u) = (u-1)^2 + 5(u-1) + 2 =$

$u^2 - 2u + 1 + 5u - 5 + 2$

$= u^2 + 3u - 2$.

Therefore, $f(x) = x^2 + 3x - 2$.

Calculating the integral: $\int_0^3 f(x)dx = \int_0^3 (x^2 + 3x - 2)dx$

$= \left[\frac{x^3}{3} + \frac{3x^2}{2} - 2x\right]_0^3 = 9 + \frac{27}{2} - 6 = \frac{33}{2}$.

12. (3) $\int \frac{1-5\cos^2 x}{\sin^5 x \cos^2 x} dx = \int \frac{dx}{\sin^5 x \cos^2 x} - 5\int \frac{dx}{\sin^5 x}$

$= \int \frac{\sec^2 x\, dx}{\sin^5 x} - 5\int \frac{dx}{\sin^5 x}$

By IBP:

$= \frac{\tan x}{\sin^5 x} - \int \left(-\frac{5}{\sin^6 x}\right) \cos x \cdot \tan x\, dx - 5\int \frac{dx}{\sin^5 x}$

$= \frac{\tan x}{\sin^5 x} + C$

So $f(x) = \frac{\tan x}{\sin^5 x}$

$f\left(\frac{\pi}{6}\right) - f\left(\frac{\pi}{4}\right) = \frac{2^5}{\sqrt{3}} - (\sqrt{2})^5 = \frac{32}{\sqrt{3}} - 4\sqrt{2} = \frac{4}{\sqrt{3}}(8 - \sqrt{6})$

13. (210) Let $I_r = \int_0^r x|\sin \pi x|\, dx$ .....(1)

Apply King Property

$= \int_0^r (r-x)|\sin \pi x|\, dx$ .....(2)

By (1) + (2)

$2I_r = \int_0^r r|\sin \pi x|\, dx \Rightarrow I_r = \frac{r}{2}\int_0^r |\sin \pi x|\, dx$

$I_1 = \frac{1}{2}\int_0^1 |\sin \pi x|\, dx = \frac{1}{2\pi}\int_0^\pi |\sin t|\, dt = \frac{1}{2\pi}(2)$

$I_2 = \frac{2}{2}\int_0^2 |\sin \pi x|\, dx = \frac{2}{2\pi}\int_0^{2\pi} |\sin t|\, dt = \frac{2}{2\pi}(4)$

$S = \sqrt{\pi \cdot \frac{1}{2\pi} \cdot 2} + \sqrt{\pi \cdot \frac{2}{2\pi} \cdot 4} + \sqrt{\pi \cdot \frac{3}{2\pi} \cdot 6} + \ldots + \sqrt{\pi \cdot \frac{20}{2\pi}(2 \cdot 20)}$

$= 1 + 2 + 3 + \ldots + 20$

$= \frac{20 \times 21}{2} = 210$

14. (64) From $\int_0^{36} f\left(\frac{tx}{36}\right) dt = 4\alpha f(x)$, substitute $u = \frac{tx}{36}$ to get $\frac{36}{x}\int_0^x f(u)du = 4\alpha f(x)$.

Differentiating: $f(x) = \frac{\alpha}{9}[f(x) + xf'(x)]$, which simplifies to $(9-\alpha)f(x) = \alpha x f'(x)$.

This gives $\frac{f'(x)}{f(x)} = \frac{9-\alpha}{\alpha x}$, integrating to $f(x) = Kx^{\frac{9-\alpha}{\alpha}}$.

For f to be a standard parabola: $\frac{9-\alpha}{\alpha} = 2 \Rightarrow \alpha = 3$.

Thus $f(x) = Kx^2$. Using point $(2,1)$: $K = \frac{1}{4}$, so $f(x) = \frac{x^2}{4}$.

At $(-4, \beta)$: $\beta = \frac{16}{4} = 4$.

Therefore $\beta^\alpha = 4^3 = 64$